Functions & Graphs

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Function transformations**

Let $f(x) = x^2$ and $g(x) = -3f(x+2) - 4 = -3(x+2)^2 - 4$. Find $g(0)$, and describe the transformation from f to g in words.

$g(0) = -3(2)^2 - 4 = -3(4) - 4 = -16$. g is f shifted left 2, reflected over the x-axis and stretched vertically by a factor of 3, then shifted down 4.

2 **[Foundational]** • **Target: Function transformations**

Using the same $g(x) = -3(x+2)^2 - 4$, find $g(-2)$, and explain why this is the vertex of g's graph.

$g(-2) = -3(0)^2 - 4 = -4$. Since $f(x) = x^2$ has its vertex (minimum) at $x = 0$, and g shifts f left by 2, the transformed vertex sits at $x = -2$.

3 **[Foundational]** • **Target: Even and odd functions**

Classify $f(x) = x^4 + 2x^2 - 5$ as even, odd, or neither.

Only even-degree terms appear (x^4, x^2, and the constant), so by Theorem 3.1 f is even. Check: $f(2) = 16 + 8 - 5 = 19$ and $f(-2) = 16 + 8 - 5 = 19$ ✓.

4 **[Foundational]** • **Target: Even and odd functions**

Explain why $k(x) = x^2 + x$ is neither even nor odd.

k mixes an even-degree term (x^2) with an odd-degree term (x), so Theorem 3.1 doesn't apply to the whole function. Checking directly: $k(1) = 2$ and $k(-1) = 0$, which is neither equal to $k(1)$ (even) nor $-k(1) = -2$ (odd).

5 **[Foundational]** • **Target: Domain restrictions**

Find the domain of $f(x) = \sqrt{2x + 6}$.

Need $2x + 6 \geq 0 \Rightarrow x \geq -3$. Domain: $x \geq -3$.

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6 [Foundational] • Target: Even and odd functions

Consider the function $f(x) = \sqrt{2x-5} + 3$. (a) Find the domain and range of $f(x)$. (b) Find the explicit equation for the inverse function $f^{-1}(x)$, including its restricted domain.

Part (a): Domain and Range of f: 1. Radicand non-negative: $2x - 5 \geq 0 \Rightarrow 2x \geq 5 \Rightarrow \text{dom}\mathbf{(f) = \left[\frac{5}{2}, \infty\right)}$. 2. Since $\sqrt{2x-5} \geq 0$, $f(x) = \sqrt{2x-5} + 3 \geq 3 \Rightarrow \text{range}\mathbf{(f) = [3, \infty)}$. Part (b): Derive $f^{-1}(x)$: 1. Set $y = \sqrt{2x-5} + 3$ and swap x and y: $x = \sqrt{2y-5} + 3$ 2. Isolate the square root: $x - 3 = \sqrt{2y-5}$ 3. Square both sides (valid for $x \geq 3$).

7 [Foundational] • Target: Domain restrictions

Let $f(x) = \frac{2x+1}{x-3}$ for $x \neq 3$. (a) Determine the inverse function $f^{-1}(x)$. (b) State the domain and range of $f^{-1}(x)$.

(a) Find the inverse function: Set $y = \frac{2x+1}{x-3}$. Swap x and y:
$x = \frac{2y+1}{y-3} \Rightarrow x(y-3) = 2y+1 \Rightarrow xy - 3x = 2y + 1$. Group y terms:
$xy - 2y = 3x + 1 \Rightarrow y(x-2) = 3x + 1 \Rightarrow \mathbf{f^{-1}(x) = \frac{3x+1}{x-2}}$. (b) Domain and Range: The denominator of $f^{-1}(x)$ cannot be zero, so the domain is $\mathbf{(-\infty, 2) \cup (2, \infty)}$ (all $x \neq 2$). The range of f^{-1} equals the domain of f, which is.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Piecewise functions

Find all vertical and horizontal asymptotes, the exact coordinates of any holes, and all intercepts of the rational function $R(x) = \frac{2x^2 + 5x - 3}{x^2 + 4x + 3}$.

Factor numerator and denominator completely: $R(x) = \frac{(2x-1)(x+3)}{(x+1)(x+3)}$. The factor $(x + 3)$ cancels for $x \neq -3$. Evaluate the simplified form $g(x) = \frac{2x-1}{x+1}$ at $x = -3$: $g(-3) = \frac{2(-3)-1}{-3+1} = \frac{-7}{-2} = 3.5$. Key features:Hole: $(-3, 3.5)$Vertical asymptote: $x + 1 = 0 \Rightarrow x = -1$Horizontal asymptote: Equal degrees (2 and 2) $\Rightarrow y = 2/1 = 2$$x$-intercept: Numerator root $2x - 1 = 0 \Rightarrow (0.5, 0)$$y$-intercept.

9 [Intermediate] • Target: Function transformations

Given $f(x) = e^x$ and $g(x) = \ln(x + 1)$, find the domain of $(g \circ f)(x)$ and evaluate $(g \circ f)(0)$.

Step 1: Formulate composite function: $(g \circ f)(x) = g(f(x)) = \ln(e^x + 1)$. Step 2: Analyze domain requirements: The argument of the natural log must be strictly positive: $e^x + 1 > 0$. Since $e^x > 0$ for all real numbers x, $e^x + 1 > 1 > 0$ holds universally. The domain is all real numbers: $\mathbf{(-\infty, \infty)}$. Step 3: Evaluate at $x = 0$: $(g \circ f)(0) = \ln(e^0 + 1) = \ln(1 + 1) = \ln\mathbf{(2)} \approx 0.693$.

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10 [Intermediate] • Target: Even and odd functions

The function $f(x) = \frac{4x-7}{2x+8}$ has horizontal asymptote $y = H$ and vertical asymptote $x = V$. Determine the value of $H + V$.

Step 1: Find horizontal asymptote $y = H$: Since degrees of numerator and denominator are both 1, the horizontal asymptote is the ratio of leading coefficients: $H = \frac{4}{2} = 2$. Step 2: Find vertical asymptote $x = V$: Set denominator to zero: $2x + 8 = 0 \Rightarrow 2x = -8 \Rightarrow V = -4$. Step 3: Compute sum: $H + V = 2 + (-4) = \mathbf{-2}$.

11 [Intermediate] • Target: Domain restrictions

The graph of $y = f(x)$ is transformed to $y = -2f(x - 3) + 5$. Describe the sequence of geometric transformations applied to the parent function.

Transformations in order: 1. Horizontal shift: Shift right by 3 units (due to $x - 3$). 2. Vertical stretch: Stretch vertically by a factor of 2. 3. Reflection: Reflect across the x-axis (due to the negative sign). 4. Vertical shift: Shift upward by 5 units.

12 [Intermediate] • Target: Piecewise functions

If $f(x) = 2x^2 - 3$ and $g(x) = \sqrt{x + 1}$, find the domain of the composite function $(g \circ f)(x)$.

Step 1: Form composition: $(g \circ f)(x) = g(f(x)) = \sqrt{(2x^2 - 3) + 1} = \sqrt{2x^2 - 2}$. Step 2: Find domain by requiring radicand ≥ 0: $2x^2 - 2 \geq 0 \Rightarrow x^2 \geq 1$. Step 3: Solve inequality: $\mathbf{(-\infty, -1] \cup [1, \infty)}$.

13 [Intermediate] • Target: Function transformations

Find the slant (oblique) asymptote of the rational function: $g(x) = \frac{2x^3 - 5x^2 + 4x + 1}{x^2 - x + 1}$

Step 1: Perform polynomial long division: Dividing $2x^3 - 5x^2 + 4x + 1$ by $x^2 - x + 1$: 1. First term: $2x^3/x^2 = 2x$. Multiply: $2x(x^2 - x + 1) = 2x^3 - 2x^2 + 2x$. Subtract: $(-5x^2 + 4x) - (-2x^2 + 2x) = -3x^2 + 2x + 1$. 2. Second term: $-3x^2/x^2 = -3$. Multiply: $-3(x^2 - x + 1) = -3x^2 + 3x - 3$. Subtract: $(2x + 1) - (3x - 3) = -x + 4$ (remainder). Step 2: Equation of slant asymptote: The quotient is the linear equation: $\mathbf{y = 2x - 3}$.

14 [Intermediate] • Target: Even and odd functions

Find the vertical and horizontal asymptotes of the rational function: $f(x) = \frac{4x^2 - 1}{2x^2 + 5x - 3}$

Step 1: Factor numerator and denominator: $f(x) = \frac{(2x-1)(2x+1)}{(2x-1)(x+3)} = \frac{2x+1}{x+3} \quad \left(x \neq \frac{1}{2}\right)$ Step 2: Vertical asymptote: The non-canceling factor in denominator is $x + 3 = 0 \Rightarrow \mathbf{x = -3}$. (The point at $x = 1/2$ is a removable hole, not a vertical asymptote). Step 3: Horizontal asymptote: Equal degrees (degree 2): ratio of leading coefficients is $y = \frac{4}{2} = \mathbf{2}$.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Domain restrictions** • **Cross-Lesson: 2.01 Sequences & Series**

The polynomial $P(x) = 2x^3 + kx^2 - 7x + 6$ has $(x - 2)$ as a factor. (a) Find the value of k. (b) Factor $P(x)$ completely over the real numbers and list all its roots.

Part (a): Find k via the Factor Theorem: By Theorem 4.1, $(x - 2)$ is a factor if and only if $P(2) = 0$: $P(2) = 2(2)^3 + k(2)^2 - 7(2) + 6 = 0$ $2(8) + 4k - 14 + 6 = 16 + 4k - 8 = 4k + 8 = 0 \Rightarrow 4k = -8 \Rightarrow \mathbf{k = -2}$. Part (b): Divide and factor completely: Polynomial division of $2x^3 - 2x^2 - 7x + 6$ by $x - 2$: $2x^3 - 2x^2 - 7x + 6 = (x - 2)(2x^2 + 2x - 3)$. Solve $2x^2 + 2x - 3 = 0$ using the quadratic formula.

16 **[Challenging]** • **Target: Piecewise functions** • **Cross-Lesson: 4.01 Matrices & Vectors**

Solve the logarithmic equation for x: $\log_3(x + 2) + \log_3(x - 4) = 3$

Step 1: Determine domain restrictions: $x + 2 > 0 \Rightarrow x > -2$ and $x - 4 > 0 \Rightarrow x > 4$. Both hold simultaneously when $\mathbf{x > 4}$. Step 2: Combine logarithms using the product property: $\log_3((x + 2)(x - 4)) = 3$ Step 3: Convert to exponential form: $(x + 2)(x - 4) = 3^3 = 27$ $x^2 - 2x - 8 = 27 \Rightarrow x^2 - 2x - 35 = 0$. Step 4: Factor the quadratic: $(x - 7)(x + 5) = 0 \Rightarrow x = 7 \quad \text{or} \quad x = -5$. Step 5: Apply domain check: $x = -5$ produces $\log_3(-3)$, which is.

17 **[Challenging]** • **Target: Function transformations** • **Cross-Lesson: 5.01 Introductory Calculus**

Solve the rational equation for all real values of x: $\frac{x}{x-2} + \frac{1}{x-4} = \frac{2}{(x-2)(x-4)}$

Step 1: Identify domain restrictions: Denominators cannot be zero: $xeq2$ and $xeq4$. Step 2: Clear denominators by multiplying by $(x - 2)(x - 4)$: $x(x - 4) + 1(x - 2) = 2$ $x^2 - 4x + x - 2 = 2 \Rightarrow x^2 - 3x - 4 = 0$. Step 3: Factor the quadratic equation: $(x - 4)(x + 1) = 0 \Rightarrow x = 4 \quad \text{or} \quad x = -1$. Step 4: Check for extraneous solutions: $x = 4$ violates the domain restriction $xeq4$ (it produces division by zero), and must be rejected. $x = -1$ is valid.

18 **[Challenging]** • **Target: Even and odd functions** • **Cross-Lesson: 6.01 Probability**

Find all vertical, horizontal, and slant asymptotes and any removable discontinuities (holes) of the rational function $f(x) = \frac{3x^2 + 1}{x^2 - 4}$. Describe its end behavior using limits.

Step 1: Domain and vertical asymptotes: The denominator factors as $x^2 - 4 = (x - 2)(x + 2)$. The numerator $3x^2 + 1 > 0$ has no real zeros. Thus, there are no common factors and no removable holes. The vertical asymptotes are $x = 2$ and $x = -2$. Step 2: Horizontal asymptote and end behavior: Numerator and denominator have equal degree $n = d = 2$: $\lim_{x \to \pm\infty} f(x) = \lim_{x \to \pm\infty} \frac{3 + 1/x^2}{1 - 4/x^2} = \frac{3}{1} = 3$. The horizontal asymptote is $y = 3$. Notice tha.

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19 [Challenging] • Target: Domain restrictions • Cross-Lesson: 1.01 Complex Numbers

Determine the value of the constant k for which the function $f(x)$ is continuous at $x = 2$:

$f(x) = \{kx^2 + 2x, , x \leq 2; 3x + k, , x > 2$

Step 1: Apply the definition of continuity at a point (Definition 6.4): Continuity requires $\lim_{x\to 2^-} f(x) = \lim_{x\to 2^+} f(x) = f(2)$. Step 2: Evaluate the one-sided limits: Left-hand limit and function value: $\lim_{x\to 2^-} f(x) = f(2) = k(2)^2 + 2(2) = 4k + 4$. Right-hand limit: $\lim_{x\to 2^+} f(x) = 3(2) + k = 6 + k$. Step 3: Set them equal and solve for k: $4k + 4 = 6 + k \Rightarrow 3k = 2 \Rightarrow \mathbf{k = \frac{2}{3}}$. Check: For $k = 2/3$.

20 [Challenging] • Target: Piecewise functions • Cross-Lesson: 2.01 Sequences & Series

Determine the slant asymptote of $g(x) = \frac{2x^3 - 3x^2 + 5}{x^2 + 1}$ and find the coordinates of any point where the curve intersects this asymptote.

Since $\deg(\text{num}) = 3$ and $\deg(\text{denom}) = 2$ with $3 - 2 = 1$, a slant asymptote exists. Perform polynomial long division: $(2x^3 - 3x^2 + 5) \div (x^2 + 1) = (2x - 3) + \frac{-2x + 8}{x^2 + 1}$. The linear quotient gives the slant asymptote: $y = 2x - 3$. To find where the graph intersects this asymptote, set the remainder equal to zero: $\frac{-2x + 8}{x^2 + 1} = 0 \Rightarrow -2x + 8 = 0 \Rightarrow x = 4$. Substitute $x = 4$ into the asymptote equation: $y = 2(4) - 3 = 5$. The curve intersects.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Function transformations

If f is an odd function and $f(2) = 13$, find $f(-2)$.

a. **-13** ✓ **b.** 13

c. -12 **d.** 2

Odd means $f(-x) = -f(x)$, so $f(-2) = -f(2) = -13$.

22 [Foundational] • Target: Even and odd functions

If f is an even function and $f(3) = -7$, find $f(-3)$.

a. **-7** ✓ **b.** 7

c. -6 **d.** 3

Even means $f(-x) = f(x)$, so $f(-3) = f(3) = -7$.

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23 [Foundational] • Target: Domain restrictions

Consider the function $f(x) = x^2 - 4x + 3$. What are the coordinates of the vertex of the parabola represented by this function?

a. **(2, -1)** ✓

b. (2, 1)

c. (1, 0)

d. (3, 0)

The given function is a quadratic function in the form $f(x) = ax^2 + bx + c$, where $a = 1$, $b = -4$, and $c = 3$. The x-coordinate of the vertex of a parabola is given by the formula $x = -\frac{b}{2a}$. Substituting the values, we get $x = -\frac{-4}{2(1)} = \frac{4}{2} = 2$. To find the y-coordinate of the vertex, we substitute this x-value back into the function: $f(2) = (2)^2 - 4(2) + 3 = 4 - 8 + 3 = -1$. Therefore, the coordinates of the vertex are (2, -1).

24 [Foundational] • Target: Piecewise functions

The graph of the function $f(x) = -x^2 + 4x - 3$ is shown below. Determine the y-coordinate of the vertex of the parabola.

a. **1** ✓

b. 2

c. 3

d. 4

The vertex of a parabola in the form $f(x) = ax^2 + bx + c$ occurs at $x = -\frac{b}{2a}$. In this case, $a = -1$ and $b = 4$. So, the x-coordinate of the vertex is $x = -\frac{4}{2(-1)} = -\frac{4}{-2} = 2$. To find the y-coordinate of the vertex, substitute this x-value back into the function: $f(2) = -(2)^2 + 4(2) - 3 = -4 + 8 - 3 = 1$. Therefore, the vertex is at (2, 1) and the y-coordinate is 1.

25 [Foundational] • Target: Function transformations

Consider the graph of the function $f(x)$ shown below. What is the value of $f(0)$?

a. **-1** ✓

b. 1

c. 0

d. 2

The value of $f(0)$ is the y-coordinate of the point where the graph of the function intersects the y-axis. From the graph, we can see that the function intersects the y-axis at the point $(0, -1)$. Therefore, $f(0) = -1$.

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26 [Foundational] • Target: Even and odd functions

The graph of the function $y = f(x)$ is shown below. What is the value of $f(1)$?

a. **0** ✓

b. 2

c. -1

d. 3

The value of $f(1)$ corresponds to the y-coordinate of the point on the graph where the x-coordinate is 1. Looking at the provided graph, the point $(1, 0)$ is marked. Therefore, $f(1) = 0$.

27 [Foundational] • Target: Domain restrictions

Describe the transformation from $f(x)$ to $g(x) = 3f(x - 5) + 7$.

a. **Shift right 5, stretch by 3, shift up 7.** ✓

b. Shift left 5, stretch by 3, shift up 7.

c. Shift right 5, stretch by 3, shift down 7.

d. Shift right 5, shift up 7 (no stretch).

$x - 5$ shifts right 5; the factor 3 stretches; $+7$ shifts up 7.

28 [Foundational] • Target: Piecewise functions

The graph of $f(x) = x^2$ is shifted to $g(x) = (x + 6)^2 - 4$. Find the vertex of g.

a. **(-6,\ -4)** ✓

b. (6,\ -4)

c. (-6,\ 4)

d. (-4,\ -6)

For $g(x) = (x - h)^2 + k$, the vertex is (h, k). Here $g(x) = (x + 6)^2 - 4$, so the vertex is $(-6, \ -4)$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Function transformations

Consider the function $f(x) = x^2 - 4x + 3$. The graph of this function is a parabola. What is the y-intercept of this parabola?

a. **3** ✓

b. -3

c. 0

d. -4

The y-intercept of a function occurs where the graph crosses the y-axis, which means the x-coordinate is 0. To find the y-intercept, substitute $x = 0$ into the function $f(x) = x^2 - 4x + 3$. $f(0) = (0)^2 - 4(0) + 3$ $f(0) = 0 - 0 + 3$ $f(0) = 3$ Therefore, the y-intercept is 3. Common errors might arise from incorrectly substituting values or misinterpreting what the y-intercept represents.

30 [Intermediate] • Target: Even and odd functions

Consider the function $f(x) = x^2 - 4x + 3$. The graph of this function is a parabola. Find the x-coordinate of the vertex of the parabola.

a. 1

b. **2** ✓

c. 3

d. -1

The given function is a quadratic function of the form $f(x) = ax^2 + bx + c$, where $a = 1$, $b = -4$, and $c = 3$. The x-coordinate of the vertex of a parabola is given by the formula $x = -\frac{b}{2a}$. Substituting the values, we get $x = -\frac{-4}{2(1)} = \frac{4}{2} = 2$. Therefore, the x-coordinate of the vertex is 2.

31 [Intermediate] • Target: Domain restrictions

Consider the function $f(x) = x^2 - 4x + 3$. What are the x-intercepts of the graph of this function?

a. **(1, 0) and (3, 0)** ✓

b. (0, 1) and (0, 3)

c. (-1, 0) and (-3, 0)

d. (1, 0) and (-3, 0)

The x-intercepts of a function occur where the function's value is zero, i.e., $f(x) = 0$. We need to solve the quadratic equation $x^2 - 4x + 3 = 0$. This equation can be factored as $(x - 1)(x - 3) = 0$. Therefore, the solutions are $x = 1$ and $x = 3$. This means the x-intercepts are $(1, 0)$ and $(3, 0)$.

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32 [Intermediate] • Target: Piecewise functions

Consider the function $f(x) = x^2 - 4x + 3$. What is the y-intercept of the graph of this function?

a. **3 ✓**

b. -3

c. 4

d. -4

The y-intercept of a function's graph is the point where the graph crosses the y-axis. This occurs when $x = 0$. To find the y-intercept, substitute $x = 0$ into the function: $f(0) = (0)^2 - 4(0) + 3$ $f(0) = 0 - 0 + 3$ $f(0) = 3$ Therefore, the y-intercept is 3.

33 [Intermediate] • Target: Function transformations

Determine whether $f(x) = x^5 + 2x$ is even, odd, or neither.

a. Even

b. **Odd ✓**

c. Neither

d. Cannot be determined

Every term has odd degree, so $f(-x) = -f(x)$: odd.

34 [Intermediate] • Target: Even and odd functions

Consider the function $f(x) = x^3 - 6x^2 + 5$. A tangent line to the curve of $f(x)$ is drawn at the point $P(a, f(a))$. This tangent line also intersects the curve at another point $Q(b, f(b))$. If the x-coordinate of the point P is $a = 1$, determine the x-coordinate of the point Q.

a. **6 ✓**

b. 5

c. 4

d. 3

The function is $f(x) = x^3 - 6x^2 + 5$. The derivative is $f'(x) = 3x^2 - 12x$. At $x = a = 1$, the slope of the tangent line is $m = f'(1) = 3(1)^2 - 12(1) = 3 - 12 = -9$. The y-coordinate of point P is $f(1) = 1^3 - 6(1)^2 + 5 = 1 - 6 + 5 = 0$. So, $P = (1, 0)$. The equation of the tangent line at P is $y - 0 = -9(x - 1)$, which simplifies to $y = -9x + 9$. To find the intersection points of the tangent line and the curve, we set the function equal to the tangent line equation: $x^3 - 6x^2 + 5 = -9x + 9$. Rearra.

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35 [Intermediate] • Target: Domain restrictions

A function f is defined for all real numbers except $x = 0$ and $x = 1$. For all values of x in its domain, f satisfies the functional equation $f(x) + f\left(\frac{x-1}{x}\right) = 1 + x$. What is the value of $f(2)$?

a. $\frac{3}{4}$ ✓

b. $\frac{5}{4}$

c. $\frac{7}{4}$

d. $\frac{9}{4}$

To find $f(2)$, substitute strategic values of x to generate a system of equations involving the cyclic transformation $g(x) = \frac{x-1}{x}$. 1. Substitute $x = 2$: $\frac{2-1}{2} = \frac{1}{2}$ $f(2) + f\left(\frac{1}{2}\right) = 1 + 2 = 3$ $---$ (Equation 1) 2. Substitute $x = \frac{1}{2}$: $\frac{\frac{1}{2}-1}{\frac{1}{2}} = \frac{-\frac{1}{2}}{\frac{1}{2}} = -1$ $f\left(\frac{1}{2}\right) + f(-1) = 1 + \frac{1}{2} = \frac{3}{2}$ $---$ (Equation 2) 3. Substitute $x = -1$.

36 [Intermediate] • Target: Piecewise functions

If $f(x) = \sqrt{x-5}$ and $g(x) = x^2 - 4$, what is the domain of $f(g(x))$?

a. $-3 \le x \le 3$

b. $x \le -3$ **or** $x \ge 3$ ✓

c. $x \ge 3$

d. All real numbers

Need $x^2 \ge 9$, so $|x| \ge 3$, giving $x \le -3$ or $x \ge 3$.

37 [Intermediate] • Target: Function transformations

A function is defined by $f(x) = x^2$ for $x < 3$ and $f(x) = -2x + 1$ for $x \ge 3$. Find $f(3)$.

a. -6

b. -4

c. 9

d. **-5** ✓

Since $x = 3 \ge 3$, use $f(x) = -2x + 1$: $f(3) = -2(3) + 1 = -5$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Even and odd functions • Cross-Lesson: 6.01 Probability

A piecewise function is defined by $f(x) = 2x + 3$ for $x < 1$ and $f(x) = x + k$ for $x \ge 1$. Find k so that f is continuous at $x = 1$.

a. 5

b. 6

c. **4** ✓

d. 1

Continuity: $2(1) + 3 = 1(1) + k \Rightarrow k = 4$.

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39 [Challenging] • Target: Domain restrictions • Cross-Lesson: 1.01 Complex Numbers

A survey was conducted on a group of 100 students regarding their preferences for two subjects, Mathematics (M) and Science (S). The results are summarized in the two-way table below, where some values are unknown. | | Science (S) | Not Science (¬S) | Total | |-------------|-------------|------------------|-------| | Mathematics (M) | 25 | x | 45 | | Not Math (¬M) | y | 30 | 55 | | Total | 55 | z | 100 | If a student is chosen at random from this group, the probability that the student prefers Mathematics but not Science, P(M ∩ ¬S), is equal to the probability that the student prefers Science but not Mathematics, P(¬M ∩ S). What is the value of x?

a. **15 ✓**

b. 20

c. 25

d. 30

The problem provides a two-way table summarizing student preferences and states that P(M ∩ ¬S) = P(¬M ∩ S). We are asked to find the value of x. From the table, we can see the following: P(M ∩ ¬S) is the probability of a student preferring Mathematics but not Science. This corresponds to the cell with value 'x' in the table. The total number of students is 100. So, P(M ∩ ¬S) = x / 100. P(¬M ∩ S) is the probability of a student preferring Science but not Mathematics. This corresponds to the cell with value 'y' in th.

40 [Challenging] • Target: Piecewise functions • Cross-Lesson: 2.01 Sequences & Series

A transformation maps point $A(1, 2)$ to $A'(-4, 2)$ and point $B(3, 5)$ to $B'(-2, 5)$. This transformation can be represented by matrix multiplication $(x\ y)' = M(x\ y) + (a\ b)$. Find the values of a and b.

a. **a = -5, b = 0 ✓**

b. a = 5, b = 0

c. a = 0, b = -5

d. a = 0, b = 5

Let the transformation be given by $(x\ y)' = M(x\ y) + (a\ b)$. We are given two points and their images under this transformation: 1. $A(1, 2) \to A'(-4, 2)$: $(-4\ 2) = M(1\ 2) + (a\ b)$ 2. $B(3, 5) \to B'(-2, 5)$: $(-2\ 5) = M(3\ 5) + (a\ b)$ Subtracting the first equation from the second, we get: $(-2 - (-4)\ 5 - 2) = M(3 - 1\ 5 - 2)$.

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41 [Challenging] • Target: Function transformations • Cross-Lesson: 4.01 Matrices & Vectors

A linear transformation in the 2D plane is represented by the matrix $M = (a, b; c, d)$. This transformation maps the point $P(1, 2)$ to $P'(-3, 7)$ and the point $Q(3, 1)$ to $Q'(5, 1)$. Determine the values of a, b, c, and d, and then calculate the sum $a + b + c + d$.

a. **9** ✓

b. 11

c. 7

d. 5

Let the linear transformation matrix be $M = (a, b; c, d)$. The transformation of a point $\mathbf{x}$ to $\mathbf{x}'$ is given by $M\mathbf{x} = \mathbf{x}'$. For point $P(1, 2)$ mapped to $P'(4, 9)$: $(a, b; c, d)(1; 2) = (4; 9)$ This yields the equations: 1) $a + 2b = 4$ 2) $c + 2d = 9$ For point $Q(3, 1)$ mapped to $Q'(7, 12)$: $(a, b; c, d)(3; 1) = (7; 12)$ This yields the equations: 3) $3a + b = 7$ 4) $3c + d = 12$ We need.

42 [Challenging] • Target: Even and odd functions • Cross-Lesson: 5.01 Introductory Calculus

A manufacturer produces light bulbs. The probability that a randomly selected light bulb is defective is p. The production process is designed such that the probability of producing exactly k defective bulbs in a batch of $n = 5$ bulbs follows the binomial probability formula: $P(X = k) = \binom{n}{k} p^k (1 - p)^{n - k}$. It is known that the probability of finding exactly 2 defective bulbs in a batch of 5 is $\frac{10}{32}$. Find the value of p.

a. 0.2

b. **0.5** ✓

c. 0.8

d. 0.75

Let X be the number of defective bulbs in a batch of $n = 5$. We are given that X follows a binomial distribution with parameters $n = 5$ and probability of success (a bulb being defective) p. The probability of exactly k defective bulbs is $P(X = k) = \binom{n}{k} p^k (1 - p)^{n - k}$. We are given that the probability of finding exactly 2 defective bulbs in a batch of 5 is $\frac{10}{32}$. Therefore, $P(X = 2) = \binom{5}{2} p^2 (1 - p)^{5 - 2} = \frac{10}{32}$. First, calculate the binomial coefficient.

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43 [Challenging] • Target: Domain restrictions • Cross-Lesson: 6.01 Probability

The graph of the function $f(x) = (x - 1)^2 - 4$ is shown in the standard (x, y) coordinate plane below. What is the range of f?

a. $[-4, \infty)$ ✓

b. $[1, \infty)$

c. $(-\infty, -4]$

d. $(-\infty, \infty)$

1. **Identify the vertex and direction of opening**: The function $f(x) = (x - 1)^2 - 4$ is written in vertex form, $f(x) = a(x - h)^2 + k$, where $a = 1$, $h = 1$, and $k = -4$. The vertex of the parabola is at $(1, -4)$. 2. **Determine the minimum output value**: Since the leading coefficient $a = 1 > 0$, the parabola opens upward, meaning that the vertex represents the absolute minimum point on the graph. The minimum value attained by $f(x)$ is -4. 3. **Determine the range**: The range represents all possible.

44 [Challenging] • Target: Piecewise functions • Cross-Lesson: 1.01 Complex Numbers

The graph of the quadratic function $f(x) = ax^2 + bx + c$ is shown in the standard (x, y) coordinate plane below, with vertex $(2, -1)$ and y-intercept $(0, 3)$. The function g is defined by $g(x) = f(x + k) - 8$, where k is a constant. If the graph of g passes through the origin $(0, 0)$, which of the following could be the value of k?

a. -5

b. -3

c. 1

d. 5 ✓

Step 1: Find the equation of $f(x)$. Using the vertex form of a parabola with vertex $(h, k_v) = (2, -1)$: $f(x) = a(x - 2)^2 - 1$ Since the graph passes through the y-intercept $(0, 3)$, substitute $x = 0$ and $y = 3$: $3 = a(0 - 2)^2 - 1 \Rightarrow 3 = 4a - 1 \Rightarrow 4a = 4 \Rightarrow a = 1$ Thus, $f(x) = (x - 2)^2 - 1$. Step 2: Apply the condition that g passes through $(0, 0)$. Since $g(0) = 0$ and $g(x) = f(x + k) - 8$: $g(0) = f(0 + k) - 8 = f(k) - 8 = 0 \Rightarrow f(k) = 8$ Step 3: Solve for k..

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45 [Challenging] • Target: Function transformations • Cross-Lesson: 2.01 Sequences & Series

Consider the transformation grid below. If point A has coordinates $(1, 2)$, what are the coordinates of its image, A', after the transformation shown?

a. **(-4, 2)** ✓

b. (-2, 1)

c. (2, -4)

d. (1, -2)

The transformation grid shows a reflection across the line $y = x$. When a point (x, y) is reflected across the line $y = x$, its image has coordinates (y, x). Given point A is $(1, 2)$, its image A' will have coordinates $(2, 1)$. However, the provided `transformation_grid` JSON shows an object at `[[1, 2], [4, 2], [2, 5]]` labeled A, B, C and its image at `[[-4, 2], [-1, 2], [-3, 5]]` labeled A', B', C'. This implies a translation, not a reflection. Let's analyze the transformation of point A. Original point.

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